ADV: Calculus (Adv), C4 Integration (Adv), Integration (Y12)

L&E Integration

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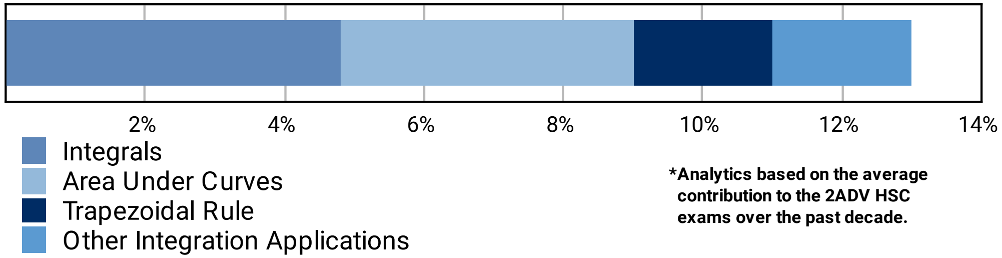
Exam Equivalent Time: 72 minutes *(based on HSC allocation of 1.5 minutes approx. per mark)*

C4 Integration



***Analytics based on the average contribution to the 2ADV HSC exams over the past decade.**

HISTORICAL CONTRIBUTION

- *C4 Integration* has contributed a substantial 13.0% to past Advanced exams, on average over the past decade.
- This topic has been split into four sub-topics for analysis purposes: *1-Integrals (4.8%), 2-Areas Under Curves (4.2%), 3-Trapezoidal Rule (2.0%) and 4-Other Integration Examples (2.0%).*
- This analysis looks at *Integrals.*

HSC ANALYSIS - What to expect and common pitfalls

- *Integrals (4.8%)* is made up of standard integration (1.2%), log and exponential integration (2.4%) and trig integration (1.2%).
- Dedicated questions testing log and exponential integration have consistently appeared in past HSC exams - absent in just 2 exams of the past decade and including a record 3 separate questions in the 2020 exam.
- Questions requiring students to differentiate then integrate have easily been the worst answered, causing surprising problems in 2020 and 2019 - an important revision area.
- Integration of a fraction that results in a logarithm is a 2-3 mark question type that has appeared 5 times in the last decade, but just once in the last 5 years (*2020 Adv 17*).
- We note *2019 13c(ii)* surprisingly produced a sub-50% mean mark and deserves attention.

Questions

1. Calculus, 2ADV C4 2014 HSC 4 MC

Which expression is equal to $\int e^{2x}\,dx$?

(A) $e^{2x}+C$

(B) $2e^{2x}+C$

(C) $\dfrac{e^{2x}}{2}+C$

(D) $\dfrac{e^{2x+1}}{2x+1}+C$

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2. Calculus, 2ADV C4 2020 HSC 4 MC

What is $\int e+e^{3x}\,dx$?

A. $ex+3e^{3x}+c$

B. $ex+\dfrac{1}{3}e^{3x}+c$

C. $e+3e^{3x}+c$

D. $e+\dfrac{1}{3}e^{3x}+c$

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3. Calculus, 2ADV C4 2012 HSC 9 MC

What is the value of $\displaystyle\int_1^4 \frac{1}{3x}\,dx$?

(A) $\dfrac{1}{3}\ln 3$

(B) $\dfrac{1}{3}\ln 4$

(C) $\ln 9$

(D) $\ln 12$

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4. Calculus, 2ADV C4 2013 HSC 11e

Find $\int e^{4x+1}dx$ *(2 marks)*

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5. Calculus, 2ADV C4 2004 HSC 3bi

Evaluate $\displaystyle\int_1^2 e^{3x}\,dx$. *(2 marks)*

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6. Calculus, 2ADV C4 2006 HSC 2bi

Find $\int 1+e^{7x}\,dx$. *(2 marks)*

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7. Calculus, 2ADV C4 2015 HSC 11h

Find $\displaystyle\int \frac{x}{x^2-3}\,dx$. *(2 marks)*

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8. Calculus, 2ADV C4 2018 HSC 11e

Evaluate $\displaystyle\int_0^3 e^{5x}\,dx$. *(2 marks)*

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9. Calculus, 2ADV C4 2016 HSC 12d

i. Differentiate $y=xe^{3x}$. *(1 mark)*

ii. Hence find the exact value of $\displaystyle\int_0^2 e^{3x}(3+9x)\,dx$. *(2 marks)*

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10. Calculus, 2ADV C4 2004 HSC 3bii

Find $\displaystyle\int \frac{x}{x^2-3}\,dx$. *(2 marks)*

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11. Calculus, 2ADV C4 2006 HSC 2bii

Evaluate $\displaystyle\int_0^3 \frac{8x}{1+x^2}\,dx$. *(3 marks)*

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12. Calculus, 2ADV C4 2019 MET1-N 3

Evaluate $\displaystyle\int_0^1 e^x-e^{-x}\,dx$. *(2 marks)*

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13. Calculus, 2ADV C4 2020 HSC 17

Find $\displaystyle\int \frac{x}{4+x^2}\,dx$. *(2 marks)*

14. Calculus, 2ADV C4 SM-Bank 1

If $m = \displaystyle\int_1^3 \frac{2}{x}\,dx$, express e^m in its simplest form. *(2 marks)*

15. Calculus, 2ADV C4 SM-Bank 3

If $f'(x) = \dfrac{x^2}{x^3+1}$ and $f(1) = \log_e 2,$ find $f(x)$. *(3 marks)*

16. Calculus, 2ADV C4 2005 HSC 2ci

Find $\displaystyle\int \frac{6x^2}{x^3+1}\,dx$. *(2 marks)*

17. Calculus, 2ADV C4 2008 HSC 2ci

Find $\displaystyle\int \frac{dx}{x+5}$. *(1 mark)*

18. Calculus, 2ADV C4 2010 HSC 2dii

Find $\displaystyle\int \frac{x}{4+x^2}\,dx$. *(2 marks)*

19. Calculus, 2ADV C4 2011 HSC 4b

Evaluate $\displaystyle\int_e^{e^3} \frac{5}{x}\,dx$ *(2 marks)*

20. Calculus, 2ADV C4 2012 HSC 12b

Find $\displaystyle\int \frac{4x}{x^2+6}\,dx$. *(2 marks)*

21. Calculus, 2ADV C4 2013 HSC 11f

Evaluate $\displaystyle\int_0^1 \frac{x^2}{x^3+1}\,dx$ *(3 marks)*

22. Calculus, 2ADV C4 2019 HSC 13c

i. Differentiate $(\ln x)^2$. *(2 marks)*

ii. Hence, or otherwise, find $\displaystyle\int \frac{\ln x}{x}\,dx$. *(1 mark)*

23. Calculus, 2ADV C4 2020 HSC 18

a. Differentiate $e^{2x}(2x+1)$. *(2 marks)*

b. Hence, find $\displaystyle\int (x+1)e^{2x}\,dx$. *(1 marks)*

Worked Solutions

1. Calculus, 2ADV C4 2014 HSC 4 MC

$$\int e^{2x}\,dx$$
$$= \frac{1}{2}e^{2x} + C$$
$$\Rightarrow\ C$$

2. Calculus, 2ADV C4 2020 HSC 4 MC

$$\int e + e^{3x} = ex + \frac{1}{3}e^{3x} + c$$

(Note e is a simple constant here)

$$\Rightarrow B$$

3. Calculus, 2ADV C4 2012 HSC 9 MC

$$\int_1^4 \frac{1}{3x}dx$$
$$= \frac{1}{3}[\ln x]_1^4$$
$$= \frac{1}{3}[\ln 4 - \ln 1]$$
$$= \frac{1}{3}\ln 4$$
$$\Rightarrow B$$

TIP: Note that $\ln(1) = 0$, as $e^0 = 1$

4. Calculus, 2ADV C4 2013 HSC 11e

$$\int e^{4x+1}\,dx = \frac{1}{4}e^{4x+1} + C$$

5. Calculus, 2ADV C4 2004 HSC 3bi

$$\int_1^2 e^{3x}\,dx = \frac{1}{3}\left[e^{3x}\right]_1^2$$
$$= \frac{1}{3}\left(e^6 - e^3\right)$$

6. Calculus, 2ADV C4 2006 HSC 2bi

$$\int 1 + e^{7x}\,dx = x + \frac{1}{7}e^{7x} + c$$

7. Calculus, 2ADV C4 2015 HSC 11h

$$\int \frac{x}{x^2-3}\,dx$$
$$= \frac{1}{2}\int \frac{2x}{x^2-3}\,dx$$
$$= \frac{1}{2}\ln\left(x^2-3\right) + C$$

8. Calculus, 2ADV C4 2018 HSC 11e

$$\int_0^3 e^{5x}\,dx = \left[\frac{1}{5}e^{5x}\right]_0^3$$
$$= \frac{1}{5}e^{15} - \frac{1}{5}e^0$$
$$= \frac{1}{5}\left(e^{15} - 1\right)$$

9. Calculus, 2ADV C4 2016 HSC 12d

i. $y = xe^{3x}$

Using product rule:

$$\frac{dy}{dx} = x \cdot 3e^{3x} + 1 \cdot e^{3x}$$
$$= e^{3x}(1+3x)$$

ii. $\displaystyle\int_0^2 e^{3x}(3+9x)\,dx$

$$= 3\int_0^2 e^{3x}(1+3x)\,dx$$
$$= 3\left[xe^{3x}\right]_0^2$$
$$= 3\left(2e^6 - 0\right)$$
$$= 6e^6$$

Worked Solutions

10. Calculus, 2ADV C4 2004 HSC 3bii

$$\begin{aligned}\int \frac{x}{x^2-3}\,dx &= \frac{1}{2}\int \frac{2x}{x^2-3}\,dx\\ &= \frac{1}{2}\ln\left(x^2-3\right)+C\end{aligned}$$

11. Calculus, 2ADV C4 2006 HSC 2bii

$$\begin{aligned}&\int_0^3 \frac{8x}{1+x^2}\,dx\\ &= 4\int_0^3 \frac{2x}{1+x^2}\,dx\\ &= 4\left[\log_e\left(1+x^2\right)\right]_0^3\\ &= 4[\log_e(1+9)-\log_e(1+0)]\\ &= 4[\log_e 10-\log_e 1]\\ &= 4\log_e 10\end{aligned}$$

12. Calculus, 2ADV C4 2019 MET1-N 3

$$\begin{aligned}\int_0^1 e^x-e^{-x}\,dx &= \left[e^x+e^{-x}\right]_0^1\\ &= \left[e+\frac{1}{e}-(1+1)\right]\\ &= e+\frac{1}{e}-2\end{aligned}$$

13. Calculus, 2ADV C4 2020 HSC 17

$$\begin{aligned}\int \frac{x}{4+x^2}\,dx &= \frac{1}{2}\int \frac{2x}{4+x^2}\,dx\\ &= \frac{1}{2}\log_e\left(4+x^2\right)+c\end{aligned}$$

14. Calculus, 2ADV C4 SM-Bank 1

$$\begin{aligned}\int_1^3 \frac{2}{x}\,dx &= [2\log_e x]_1^3\\ m &= 2\log_e 3-2\log_e 1\\ &= 2\log_e 3\\ \therefore e^m &= e^{2\log_e 3}\\ &= e^{\log_e 9}\\ &= 9\end{aligned}$$

15. Calculus, 2ADV C4 SM-Bank 3

$$\begin{aligned}f(x) &= \int f'(x)\,dx\\ &= \int \frac{x^2}{x^3+1}\,dx\\ &= \frac{1}{3}\int \frac{3x^2}{x^3+1}\,dx\\ &= \frac{1}{3}\log_e\left|\left(x^3+1\right)\right|+c\end{aligned}$$

$$\text{Since }\ f(1)=\log_e 2,$$

$$\begin{aligned}\frac{1}{3}\log_e 2+c &= \log_e 2\\ c &= \frac{2}{3}\log_e 2\end{aligned}$$

$$\therefore\ f(x)=\frac{1}{3}\log_e\left(x^3+1\right)+\frac{2}{3}\log_e 2$$

16. Calculus, 2ADV C4 2005 HSC 2ci

$$\begin{aligned}\int \frac{6x^2}{x^3+1}\,dx &= 2\int \frac{3x^2}{x^3+1}\,dx\\ &= 2\ln\left(x^3+1\right)+C\end{aligned}$$

17. Calculus, 2ADV C4 2008 HSC 2ci

$$\int \frac{dx}{x+5}$$
$$= \ln(x+5) + C$$

18. Calculus, 2ADV C4 2010 HSC 2dii

$$\int \frac{x}{4+x^2}\,dx$$
$$= \frac{1}{2}\int \frac{2x}{4+x^2}\,dx$$
$$= \frac{1}{2}\ln\left(4+x^2\right) + C$$

IMPORTANT: Minimise errors by adjusting the integral to fit the form $\frac{f'(x)}{f(x)}$ *before* integrating.

19. Calculus, 2ADV C4 2011 HSC 4b

$$\int_e^{e^3} \frac{5}{x}\,dx$$
$$= 5\int_e^{e^3} \frac{1}{x}\,dx$$
$$= 5[\ln x]_e^{e^3}$$
$$= 5\left(\ln e^3 - \ln e\right)$$
$$= 5(3-1)$$
$$= 10$$

MARKER'S COMMENT: Most common error was $\ln(5x)$. Minimize errors by getting the integral in the form of $\frac{f'(x)}{f(x)}$ *before* integrating.

20. Calculus, 2ADV C4 2012 HSC 12b

$$\int \frac{4x}{x^2+6}\,dx$$
$$= 2\int \frac{2x}{x^2+6}\,dx$$
$$= 2\ln\left(x^2+6\right) + C$$

21. Calculus, 2ADV C4 2013 HSC 11f

$$\int_0^1 \frac{x^2}{x^3+1}dx$$
$$= \frac{1}{3}\int_0^1 \frac{3x^2}{x^3+1}dx$$
$$= \frac{1}{3}\left[\ln\left(x^3+1\right)\right]_0^1$$
$$= \frac{1}{3}(\ln 2 - \ln 1)$$
$$= \frac{1}{3}\ln 2$$

TIP: Note that $\ln(1)=0$, because $e^0=1$

22. Calculus, 2ADV C4 2019 HSC 13c

i. $y = (\ln x)^2$

$$\frac{dy}{dx} = 2\cdot\frac{1}{x}\cdot\ln x$$
$$= \frac{2\ln x}{x}$$

ii. $$\int \frac{\ln x}{x}\,dx = \frac{1}{2}\int \frac{2\ln x}{x}dx$$
$$= \frac{1}{2}(\ln x)^2 + C$$

♦ *Mean mark part (ii) 49%.*

23. Calculus, 2ADV C4 2020 HSC 18

a. $y = e^{2x}(2x+1)$

$$\begin{aligned}\frac{dy}{dx} &= 2x^{2x}(2x+1) + 2e^{2x}\\ &= 2e^{2x}(2x+2)\\ &= 4e^{2x}(x+1)\end{aligned}$$

♦ *Mean mark part (b) 40%.*

b. $$\begin{aligned}\int (x+1)e^{2x}\,dx &= \frac{1}{4}\int 4e^{2x}(x+1)\\ &= \frac{1}{4}e^{2x}(2x+1) + c\end{aligned}$$